

Local degree conditions and matching tails for the size Ramsey numbers of star forests

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Abstract

The Burr–Erdős–Faudree–Rousseau–Schelp conjecture expresses the size Ramsey number of two star forests as a sum of antidiagonal maxima. We prove the conjectured formula under a local gap condition, and for additional families with unrestricted matching tails. In particular, for all $a, b \geq 3$, $r \geq 1$, and $u, v \geq 0$,

$$\widehat{r}(rK_{1,a} \sqcup uK_2, K_{1,b} \sqcup vK_2) = r(a + b - 1) + bu + av - (\min(a, b) - 1) \min(u, v).$$

When a, b have opposite parity, we also allow repeated nontrivial stars on both sides in an explicit multiplicity range. The proofs combine the simultaneous-suffix lower-bound framework previously recorded by rickyc with equality-host degree bounds, matchings covering prescribed vertices, and a count of the excess contributed by isolated complete components. All added proofs are symbolic. The general conjecture remains unresolved.

Scope and review. This revised manuscript presents partial results for Erdős problem 561. AI assistance was used in developing and writing it. The coordinating human reported reviewing the original local-criterion manuscript; the additions in this revision have received internal mathematical checks but have not yet received that human review or external peer review. Literature priority of the additional results is not established.

Revision. Sections 5–6 add the matching arguments and exact families with unrestricted matching tails. The local criterion and its earlier consequences are retained. This PDF revision makes no claim to settle the full problem.

1 Statement

All graphs are finite and simple. Copies need not be induced. Write $G \rightarrow (F, H)$ if every red/blue coloring of $E(G)$ contains a red copy of F or a blue copy of H . The size Ramsey number $\widehat{r}(F, H)$ is the minimum of $e(G)$ over these hosts.

Let $A = (a_1, \dots, a_s)$ and $B = (b_1, \dots, b_t)$ be nonincreasing sequences of positive integers, and put

$$F(A) = \bigsqcup_{i=1}^s K_{1,a_i}, \quad F(B) = \bigsqcup_{j=1}^t K_{1,b_j}, \quad q = s + t.$$

For $2 \leq k \leq q$, define

$$\ell_k = \max_{\substack{1 \leq i \leq s, 1 \leq j \leq t \\ i+j=k}} (a_i + b_j - 1), \quad S_k = \sum_{h=k}^q \ell_h.$$

The conjecture of Burr et al. [1] asserts $\widehat{r}(F(A), F(B)) = S_2$.

Call an index k *good* if at least one pair (i, j) attaining ℓ_k has either both a_i, b_j odd, or $\min(a_i, b_j) = 1$. The word *good* refers only to this explicit arithmetic condition.

Theorem 1 (Local criterion). *Suppose that every index $2 \leq k < q$ which is not good satisfies at least one of the following conditions:*

1. $\ell_k - \ell_{k+1} \geq 3$;

2. $\ell_k - \ell_{k+1} = 2$, and either

$$k + 1 < q \text{ and } \ell_{k+1} - \ell_{k+2} \geq 2, \quad \text{or} \quad k + 1 = q \text{ and } \ell_q \geq 2.$$

There is no condition on the final index q . Then

$$\widehat{r}(F(A), F(B)) = S_2.$$

Every host attaining this value has maximum degree at most ℓ_2 . If $q > 2$ and $\ell_2 - \ell_3 \geq 2$, it has at most one vertex of degree ℓ_2 . For $q = 2$, the last assertion holds when $\ell_2 \geq 2$.

The all-odd hypothesis in Davoodi et al. [3] makes every index good. Theorem 1 also allows mixed parity and equal successive values of ℓ_k at good indices. Its conditions involve the maximizing pairs and adjacent values, not the total number of components. Single-star cases, including a star against two arbitrary positive stars, also appear in Lortz and Mengersen [7, Theorem 7].

A forum contribution by rickyc dated 6 August 2026 [10] already gives the simultaneous-suffix deletion framework, the same distinction between good and other diagonals, and the general lower bound

$$\widehat{r}(F(A), F(B)) \geq S_2 - \sum_{k=2}^{q-1} \eta_k, \quad \eta_k = \begin{cases} 0, & k \text{ is good,} \\ 1, & k \text{ is not good.} \end{cases}$$

We do not claim these ingredients as new. The candidate addition here is the equality-host degree argument in Section 3: under the local gap conditions it removes all of these deficits. The added matching-tail arguments are given in Sections 5–6; their overlap with prior results is discussed at the end of the manuscript.

2 Deletion and degree bounds

For $1 \leq i \leq s$ and $1 \leq j \leq t$, let

$$A_i = (a_i, \dots, a_s), \quad B_j = (b_j, \dots, b_t).$$

A graph has property $\mathcal{P}_k$ if

$$G \rightarrow (F(A_i), F(B_j)) \quad \text{for every feasible } i + j = k.$$

In particular, $\mathcal{P}_2$ is the original Ramsey property. Property $\mathcal{P}_q$ concerns the two single stars K_{1,a_s} and K_{1,b_t} .

Lemma 2 (Simultaneous suffix deletion). *If G has $\mathcal{P}_k$ and $k < q$, then $G - v$ has $\mathcal{P}_{k+1}$ for every vertex v .*

This formulation and its proof already appear in [10]; we include the argument to make the presentation self-contained.

Proof. First suppose $G \rightarrow (F(A_i), F(B_j))$ and $i < s$. Suppose an edge coloring of $G - v$ avoids red $F(A_{i+1})$ and blue $F(B_j)$. Extend it by coloring all edges at v red. A red $F(A_i)$ in the extension would have at most one component meeting v . The other components contain $F(A_{i+1})$, since removing any one star from a nonincreasing list leaves a forest containing the list with its largest star removed. A blue target is unchanged. Thus no such avoiding coloring exists. The analogous assertion holds on deleting the largest blue star.

For each feasible pair (i, j) with $i + j = k + 1$, at least one of $(i - 1, j)$ and $(i, j - 1)$ is feasible. Apply the corresponding assertion to that predecessor. $\square$

We use Vizing's theorem and the following result of Fournier: if the subgraph induced by the maximum-degree vertices of a graph is a forest, the graph has a proper edge coloring with $\Delta(G)$ colors; see [4]. We call this induced subgraph the maximum-degree core.

Lemma 3. *If G has $\mathcal{P}_k$, then $\Delta(G) \geq \ell_k - 1$. If k is good, then $\Delta(G) \geq \ell_k$.*

Proof. Choose a pair (i, j) attaining ℓ_k , and write $a = a_i$, $b = b_j$. If $\Delta(G) \leq a + b - 3$, Vizing gives at most $a + b - 2$ matching color classes. Merge at most $a - 1$ classes into red and at most $b - 1$ into blue. This avoids both largest stars and therefore avoids both target forests, a contradiction.

If a, b are odd, a graph of maximum degree at most $a + b - 2$ also admits a coloring with red maximum degree at most $a - 1$ and blue maximum degree at most $b - 1$. Indeed, embed it in an $(a + b - 2)$ -regular simple graph, factor that even-regular graph into 2-factors, and put $(a - 1)/2$ factors in red and $(b - 1)/2$ in blue. Restrict to the original graph. The zero-degree case is immediate. This is the parity argument in [3, Lemma 2.1]. A regular supergraph can be obtained by repeatedly taking two disjoint copies and joining corresponding deficient vertices.

If $a = 1$, the all-blue coloring requires a copy of the blue forest, hence degree at least $b = \ell_k$. The case $b = 1$ is symmetric. Applying either argument to a witnessing maximizing pair proves the last assertion. $\square$

Lemma 4 (Upper construction). *The disjoint union $\bigsqcup_{h=k}^q K_{1, \ell_h}$ has $\mathcal{P}_k$.*

Proof. Fix a feasible starting pair (i, j) with $i + j = k$. At the star with ℓ_{i+j} edges, the center has at least a_i red edges or at least b_j blue edges. Select the corresponding monochromatic star, and increase i or j , respectively. Continue on the next host component. One list is exhausted in at most $q - k + 1$ steps. The selected stars in that color form its target forest, and are vertex-disjoint because they came from different host components. $\square$

3 Proof of the local criterion

Proof of Theorem 1. We prove, by decreasing induction on k , three assertions:

1. every graph with $\mathcal{P}_k$ has at least S_k edges;
2. every such graph with exactly S_k edges has maximum degree at most ℓ_k ;
3. an equality graph has at most one degree- ℓ_k vertex if $k < q$ and $\ell_k - \ell_{k+1} \geq 2$, or if $k = q$ and $\ell_q \geq 2$.

For $k = q$ this is the elementary formula

$$\widehat{r}(K_{1,a_s}, K_{1,b_t}) = a_s + b_t - 1 = \ell_q.$$

For completeness, any graph with at most $a_s + b_t - 2$ edges can be colored with at most $a_s - 1$ red edges and at most $b_t - 1$ blue edges. The opposite inequality follows from a star with $a_s + b_t - 1$ edges. An equality graph has maximum degree at most its edge count. If two vertices had degree $\ell_q \geq 2$, their incident edges would number at least $2\ell_q - 1 > \ell_q$, which is impossible.

Now let $k < q$ and assume the assertions for all larger indices. Take a graph G with $\mathcal{P}_k$, a maximum-degree vertex v , and set $D = \Delta(G)$, $H = G - v$. By Lemma 2, H has $\mathcal{P}_{k+1}$, so

$$e(G) = D + e(H) \geq D + S_{k+1}.$$

If k is good, Lemma 3 gives $D \geq \ell_k$ and the lower bound follows. Otherwise suppose, for a contradiction, that $e(G) < S_k$. Lemma 3 and integrality force

$$e(G) = S_k - 1, \quad D = \ell_k - 1, \quad e(H) = S_{k+1}. \quad (1)$$

The inductive equality bound gives $\Delta(H) \leq \ell_{k+1}$.

If $\ell_k - \ell_{k+1} \geq 3$, every vertex other than v has degree in G at most $\ell_{k+1} + 1 \leq D - 1$. The maximum-degree core consists of v alone.

In the remaining allowed case, $\ell_k - \ell_{k+1} = 2$. Every degree- D vertex other than v has degree exactly $\ell_{k+1} = D - 1$ in H and is adjacent to v . By the stated additional condition and the inductive third assertion, there is at most one such vertex. Consequently the maximum-degree core of G is a vertex or an edge.

In either case Fournier's theorem yields a proper edge coloring with D matching classes. For a pair (i, j) attaining ℓ_k , equation (1) says $D = a_i + b_j - 2$. Merge $a_i - 1$ classes into red and $b_j - 1$ into blue. Neither color contains its largest target star. This contradicts $\mathcal{P}_k$, proving the lower bound.

It remains to prove the equality assertions at index k . If $e(E) = S_k$ and E has $\mathcal{P}_k$, deleting any vertex w and applying the lower bound at $k + 1$ gives

$$d_E(w) \leq S_k - S_{k+1} = \ell_k.$$

Suppose $\ell_k - \ell_{k+1} \geq 2$ and w, u are distinct degree- ℓ_k vertices. Then $E - w$ has $\mathcal{P}_{k+1}$ and exactly S_{k+1} edges, so its maximum degree is at most ℓ_{k+1} . But u has degree at least $\ell_k - 1 > \ell_{k+1}$ there, a contradiction. This completes the induction. Lemma 4 supplies the matching upper bound. $\square$

4 Consequences and comparison

Corollary 5 (Decreasing maxima). *If $\ell_k - \ell_{k+1} \geq 2$ for every $2 \leq k < q$, the conjectured formula holds.*

Proof. If $\ell_q \geq 2$, every nonterminal index satisfies one of the local conditions. If $\ell_q = 1$, then $a_s = b_t = 1$. Every feasible pair on the penultimate diagonal has one component of size 1, so $q - 1$ is good. Every earlier index again satisfies a local condition. For $q = 2$, the single-star formula applies directly. $\square$

Corollary 6 (An even largest component). *Suppose $s, t \geq 2$, all components except a_1 have odd size, and a_1 is even. If*

$$a_1 - a_2 \geq 3, \quad b_1 - b_2 > a_1 - a_2,$$

then the conjectured formula holds provided, for every $3 \leq k \leq \min(q, t+1)$, some maximizing pair on diagonal k has $i \geq 2$.

Proof. The additional maximizing-pair condition makes every index after the first good; for $k > t+1$, row $i = 1$ is unavailable. On diagonal 3, the strict gap inequality yields

$$\ell_3 = a_2 + b_1 - 1, \quad \ell_2 - \ell_3 = a_1 - a_2 \geq 3.$$

Thus the only possibly bad nonterminal index satisfies the first local condition. $\square$

For $t = 2$, the extra maximizing-pair condition in Corollary 6 is automatic from its strict gap inequality. This already gives unrestricted lengths and repetitions in the tail of A .

Corollary 7 (A family with two independent multiplicities). *For all positive integers r, m ,*

$$\hat{r}(K_{1,6} \sqcup rK_{1,3} \sqcup mK_2, K_{1,5} \sqcup K_2) = 11 + 7r + 5m.$$

Proof. The antidiagonal maxima, in order, are

$$10, \underbrace{7, \dots, 7}_{r \text{ times}}, \underbrace{5, \dots, 5}_{m \text{ times}}, 1.$$

The first index is not good and is followed by a drop of 3. Every later maximum is attained by an odd-odd pair. Theorem 1 gives the sum displayed in the statement. $\square$

Another example, with both components of the second forest larger than one, is

$$\hat{r}(K_{1,6} \sqcup K_{1,3} \sqcup rK_2, K_{1,7} \sqcup K_{1,3}) = 24 + 7r \quad (r \geq 1).$$

Here the maxima are 12, 9, followed by r copies of 7, and finally 3. Again only the first index is bad, with a drop of 3.

Corollary 8 (Arbitrarily many even components). *Let $h, m, r_1, \dots, r_h$ be positive integers. Construct A by concatenating the blocks*

$$\left(4j + 2, \underbrace{4j - 1, \dots, 4j - 1}_{r_j \text{ entries}} \right) \quad \text{for } j = h, h-1, \dots, 1,$$

and then appending m entries equal to 1. For $B = (5, 1)$,

$$\hat{r}(F(A), F(B)) = 2h^2 + 8h + \sum_{j=1}^h r_j(4j + 3) + 5m + 1.$$

This value exceeds the lower bound obtained by rounding every even component of A down by one and applying the all-odd theorem by exactly h .

Proof. Consecutive entries of A differ by at most 3. With $B = (5, 1)$, every diagonal except the last is therefore maximized by its column-1 pair: the competing column-2 value is the preceding entry of A , whereas the column-1 value is the current entry plus 4. The maxima are the entries of A plus 4, followed by 1.

The only bad indices correspond to the h even entries $4j + 2$. Each is followed by $4j - 1$, so its maximum drops by exactly 3. All other indices are good. Summing the maxima gives

$$\sum_{j=1}^h (4j + 6) + \sum_{j=1}^h r_j (4j + 3) + 5m + 1,$$

which is the claimed expression. If each even entry is reduced by one, the resulting sequence is still nonincreasing and all its adjacent gaps are at most 2. Column 1 still supplies every nonfinal maximum; exactly h of those maxima have decreased by one. Both forests now have only odd components, so their size Ramsey number is the corresponding sum by the published all-odd theorem. Monotonicity therefore gives precisely the claimed comparison lower bound. $\square$

What the comparison establishes. The families in Corollaries 7 and 8 have neither forest uniform, contain an even component, and have more than one nontrivial component in the first forest. They therefore fall outside the uniform, all-odd, one-star, and equal-two-star hypotheses listed in [3], as well as the one-nontrivial-star-per-forest scope of Cheng’s thesis [5]. Their final maximum is 1, so they fail the Győri–Schelp numerical sufficient condition as restated in [3, Theorem 1.4]. For the example with $B = (7, 3)$, that numerical condition fails already at the first 7 when $r \geq 3$, because $\binom{7}{2} = 21 < 7r + 3$.

For Corollary 7, the bound in [10] is one edge below the exact value proved here. For Corollary 8, it is h edges below, since exactly h nonfinal diagonals are not good. Thus the growing improvement also holds relative to that specific prior forum bound, not only to the direct all-odd rounding comparison.

These observations establish noncoverage by those stated hypotheses, not priority over all results or consequences in the literature. The full original Győri–Schelp article and the full Cheng thesis were not accessible in this audit. The proof of Theorem 1 does not depend on either source. A separate novelty audit records the accessible papers and the limitations of the search.

5 Matching tools

For a prescribed set $W \subseteq V(G)$, let $d_W(G)$ be the minimum number of vertices of W left uncovered by a matching of G . For $Z \subseteq V(G)$, write $q_W(G - Z)$ for the number of odd components of $G - Z$ whose vertices all belong to W . We first derive the matching statements used below from Tutte’s standard theorems.

Lemma 9 (Prescribed matching deficiency). *For every finite graph G and $W \subseteq V(G)$,*

$$d_W(G) = \max_{Z \subseteq V(G)} (q_W(G - Z) - |Z|).$$

Proof. First we prove the zero-deficiency criterion: a matching covers W if and only if $q_W(G - Z) \leq |Z|$ for every Z . Necessity follows because covering an odd all- W component requires a matching edge to Z , and different such components require different vertices of Z .

For sufficiency, put $n = |V(G)|$ and $U = V(G) \setminus W$. Adjoin n new vertices N and add all missing edges within $U \cup N$, leaving all other adjacencies unchanged. The resulting graph G^+ has even order $2n$. A matching of G covering W extends to a perfect matching of G^+ by pairing the remaining vertices in the clique $U \cup N$. Conversely, deleting the added edges from a perfect matching of G^+ leaves a matching covering W , since no added edge meets W .

If G^+ has no perfect matching, Tutte's theorem [8] gives a set T such that $G^+ - T$ has more than $|T|$ odd components. Their number has the same parity as $|T|$, so it is at least $|T| + 2$. At most one component meets the surviving clique $U \cup N$. Hence at least $|T| + 1$ odd components consist entirely of vertices of W . They are unchanged components of $G - Z$, where $Z = T \cap V(G)$, contradicting $q_W(G - Z) \leq |Z|$. The empty graph causes no exception.

Now put $d = \max_Z(q_W(G - Z) - |Z|) \geq 0$. Every matching misses at least $q_W(G - Z) - |Z|$ vertices of W , giving one inequality. For the other, adjoin d new vertices, each adjacent to every vertex of W , and continue to prescribe only the old set W . After deleting a set T , if a new vertex remains, every surviving W -vertex lies in a component containing that vertex; there are no all- W components. If all new vertices were deleted, write $Z = T \cap V(G)$; there are at most $|Z| + d = |T|$ counted components by the definition of d . The zero-deficiency criterion therefore supplies a matching covering W in the enlarged graph. Discarding its edges at new vertices misses at most d vertices of W , proving the formula. $\square$

For $\Delta(G) \leq D$, write

$$C_D(G) = G[\{v : d_G(v) = D\}], \quad d_D(G) = d_{V(C_D(G))}(G).$$

Thus $C_D(G)$ may be empty. The following splitting lemma concerns degrees in the two colors, not a proper edge coloring of G .

Lemma 10 (Capacity splitting). *Let $D \geq 2$ and $\Delta(G) \leq D$. If a matching covers every vertex of $C_D(G)$, then for every integer $0 \leq p \leq D$ the edges of G can be partitioned into graphs of maximum degrees at most p and $D - p$.*

Proof. We use the multigraph form of Tutte's f -factor theorem, including nonnegative demands; see [9, Theorem 1.1]. For a constant integer demand $h \geq 0$, let $q_h(X, Y)$ count the components Q of a loopless multigraph $J - X - Y$ for which $h|Q| + e_J(Q, Y)$ is odd. An h -factor exists exactly when, for every pair of disjoint vertex sets X, Y ,

$$\Phi_h(X, Y) := h(|X| - |Y|) + \sum_{v \in Y} d_{J-X}(v) - q_h(X, Y) \geq 0.$$

All degrees here count edge multiplicities.

If J has factors of degrees $h_0 \leq h_1$ of the same parity, it has every factor degree h of that parity between them. Indeed $q_{h_0} = q_h = q_{h_1}$ for each X, Y , and Φ_h is the corresponding convex combination of Φ_{h_0} and Φ_{h_1} . The displayed criterion proves the assertion. Consequently a D -regular loopless multigraph with a perfect matching has a factor of every degree from 0 to D : it has factors of degrees 0, 1, $D - 1$, D , and the two parity intervals between these endpoints cover all possibilities.

Let M cover $C_D(G)$. Take two disjoint copies of G and, for each original vertex v , join its two copies by $D - d_G(v)$ parallel edges. The resulting loopless multigraph J is D -regular. The two copies of M , together with one cross edge for every vertex missed by M , form a perfect matching: each missed vertex has degree less than D and therefore has an available cross edge. Take a p -factor of J , color it red and its complement blue, and restrict to the original edges in one copy of G . The required degree bounds follow. $\square$

Lemma 11 (Largest-star multiplicities). *Let $a, b \geq 2$, $r, s \geq 1$, $u, v \geq 0$, and $D = a + b - 2$. If $\Delta(G) \leq D$ and*

$$G \rightarrow (rK_{1,a} \sqcup uK_2, sK_{1,b} \sqcup vK_2),$$

then $d_D(G) \geq \max(r, s)$.

Proof. Let U be the set of all vertices of $C_D(G)$ missed by a matching, and suppose $|U| < r$. None of its edges meets U . Its restriction to $G - U$ covers every vertex of degree D there. By Lemma 10, color $G - U$ with red degree at most $a - 1$ and blue degree at most $b - 1$. Color all edges incident with U red. Blue has no $K_{1,b}$, while every red $K_{1,a}$ meets U ; thus there cannot be r vertex-disjoint red copies. This contradicts the arrow property. Interchanging the colors proves the bound by s as well. $\square$

Lemma 12 (Independence in the degree core). *Let $\Delta(G) \leq D$, $C = C_D(G)$, and $d = d_D(G)$.*

1. *If $D \geq 3$ is odd, then*

$$\alpha(C) \geq 2 \left\lceil \frac{Dd}{D-1} \right\rceil.$$

2. *If $D \geq 2$ is even and z is the number of isolated connected components of G isomorphic to K_{D+1} , then*

$$\alpha(C) \geq 2d - z.$$

Every maximal independent set of C contains exactly one vertex of each such clique, and its deletion leaves z isolated K_D components.

Proof. The inequalities are immediate when $d = 0$. Otherwise choose Z attaining Lemma 9. There are $|Z| + d$ odd all-core components of $G - Z$. For such a component Q , its boundary size is

$$b(Q) = D|Q| - 2e(Q), \quad \sum_Q b(Q) \leq D|Z|.$$

Call a counted component low when $b(Q) < D$. For any order $1 \leq N \leq D$, simplicity gives

$$b(Q) \geq N(D - N + 1) \geq D,$$

where the last inequality follows from $N(D - N + 1) - D = (N - 1)(D - N) \geq 0$.

If D is odd, every counted boundary is positive and odd. Writing l for the number of low components, we obtain

$$D(|Z| + d) - (D - 1)l \leq D|Z|,$$

so $l \geq \lceil Dd/(D - 1) \rceil$. Each low component has odd order at least $D + 2$, hence is noncomplete and contains two nonadjacent vertices. Pairs from different components are mutually nonadjacent, proving the first assertion.

If D is even, nonnegative boundaries instead give $D(|Z| + d - l) \leq D|Z|$, hence $l \geq d$. A low component has order at least $D + 1$. If complete, it is K_{D+1} and all its vertices already have D internal neighbors, so its boundary is empty and it is an isolated component of G . Choose an independent pair in every other low component and one vertex in each complete one. This gives $\alpha(C) \geq 2l - z \geq 2d - z$. Maximality of an independent set forces exactly one selected vertex in each isolated K_{D+1} , proving the final assertion. $\square$

6 Unrestricted matching tails with repeated leading stars

We retain the simultaneous-suffix notation $\mathcal{P}_k$ and the definition of good indices from Sections 1 and 2. One consequence of Lemmas 2 and 3 will be used repeatedly: if all indices from k onward are good, every graph with $\mathcal{P}_k$ has at least S_k edges. To see this directly, repeatedly delete a maximum-degree vertex. At each level the removed degree is at least its diagonal maximum; summing the

removed degrees proves the bound. After the last level there is no requirement and the remaining edge count is nonnegative. This also establishes the bound for any all-good suffix, including suffixes reached after several vertex deletions.

Lemma 13 (Parity excess of isolated cliques). *Suppose H has a simultaneous-suffix requirement all of whose remaining diagonal entries are good and belong to $\{1, 2, 4, 6, \dots\}$. Let T be their sum. If H has z specified isolated K_D components, where $D \geq 4$ is even, then*

$$e(H) - T \geq z(D/2 - 1).$$

The assertion also holds after the suffix is exhausted, when $T = 0$ and no Ramsey requirement remains.

Proof. Delete a maximum-degree vertex for each remaining suffix level. If d_i is its current degree and τ_i the corresponding diagonal maximum, then $d_i \geq \tau_i$. Writing H_* for the final remainder gives

$$e(H) - T = \sum_i (d_i - \tau_i) + e(H_*),$$

with every term nonnegative. Within a specified isolated clique, the successive deleted degrees are $D - 1, D - 2, \dots, 1$, possibly stopping early. There are $D/2 - 1$ odd degrees at least three in this list. Whenever one is deleted, it exceeds the current even-or-one value τ_i by at least one. Charge one unit to that clique. If j vertices of the clique remain at the end, its $\binom{j}{2}$ remaining edges pay for all the still uncharged odd degrees in $j - 1, \dots, 1$. Distinct specified cliques use distinct deletion terms and disjoint remaining edges. This pays all the required charges. For an exhausted suffix, all charges are paid by its edges. $\square$

For $a, b \geq 3$, $r, s \geq 1$ and $u, v \geq 0$, set

$$A = (a^r, 1^u), \quad B = (b^s, 1^v), \quad m = \min(a, b), \quad c = \max(a, b), \quad D = a + b - 2, \quad n = r + s - 1.$$

Here exponents in a list denote repetitions. The diagonal profile consists of n initial values $D + 1$, followed by the good tail

$$\begin{cases} (c^v, b^{u-v}, 1^v), & u \geq v, \\ (c^u, a^{v-u}, 1^u), & v \geq u. \end{cases} \quad (2)$$

Indeed, after the leading rectangle an a -star can pair with a matching edge for exactly v more diagonals, and a b -star for exactly u more. When neither is available, only two matching edges can be paired. Thus the tail has length $u + v$, each value is at most c , and its sum is

$$T = bu + av - (m - 1) \min(u, v). \quad (3)$$

Every tail maximum has a maximizing pair containing a one, so all its indices are good. The conjectured sum is

$$S = n(D + 1) + T.$$

Lemma 14 (The complete tail bounds core independence). *Suppose $G \rightarrow (F(A), F(B))$, $\Delta(G) \leq D$, and $e(G) = S - 1$. Then*

$$\alpha(C_D(G)) \leq n + \left\lfloor \frac{n - 1}{m - 2} \right\rfloor.$$

Proof. It suffices to consider an independent core set J of size $n + t$, where $t \geq 1$. Its vertices have degree D and no edges between them, so

$$e(G - J) = n - 1 + T - Dt.$$

Write the tail as $\tau_1, \dots, \tau_L$, where $L = u + v$. If $t < L$, repeated suffix deletion and the all-good lower bound give

$$e(G - J) \geq T - \sum_{i=1}^t \tau_i.$$

Hence $Dt \leq n - 1 + ct$. If $t \geq L$, use instead $e(G - J) \geq 0$ and $T \leq cL \leq ct$, with the same conclusion. Since $D - c = m - 2 \geq 1$, the asserted bound follows. The latter case includes an empty tail and deletion beyond its end. $\square$

Theorem 15 (Opposite-parity leading stars). *Let $a, b \geq 3$ have opposite parity. If*

$$\min(r, s) \leq \left\lfloor \frac{3 \min(a, b) - 5}{2} \right\rfloor, \quad r, s \geq 1, \quad u, v \geq 0,$$

then

$$\widehat{r}(rK_{1,a} \sqcup uK_2, sK_{1,b} \sqcup vK_2) = (r + s - 1)(a + b - 1) + bu + av - (\min(a, b) - 1) \min(u, v).$$

Proof. The profile calculation and Lemma 4 give the upper bound. Fix a, b, u, v and suppose a lower-bound failure exists in the stated multiplicity region. Choose one with $r + s$ least, and pad a deficient host by disjoint edges to obtain $e(G) = S - 1$.

We first show $\Delta(G) \leq D$. If $r = s = 1$, deletion of any vertex leaves the complete good tail, of lower bound T , so every degree is at most $S - 1 - T = D$. If the tail is empty, the same inequality follows from $e(G) = D$. If $r \geq 2$, the one-color extension in the proof of Lemma 2 shows that $G - v$ arrows $((r - 1)K_{1,a} \sqcup uK_2, sK_{1,b} \sqcup vK_2)$. That preceding pair is exact by minimality, since reducing either positive multiplicity preserves the stated region. Its conjectured sum is $S - (D + 1)$, giving the same degree bound. The case $r = 1, s \geq 2$ is symmetric. This handles the base, including $m = 3$, without an additional gap assumption.

Put $p = \min(r, s)$ and $q' = \max(r, s)$. By Lemmas 11, 12 and 14,

$$2 \left\lceil \frac{Dq'}{D - 1} \right\rceil \leq \alpha(C_D(G)) \leq p + q' - 1 + \left\lfloor \frac{p + q' - 2}{m - 2} \right\rfloor.$$

The right side is at most $2q' + 1$. Indeed, after subtracting $2q'$, the expression

$$p - q' - 1 + \left\lfloor \frac{p + q' - 2}{m - 2} \right\rfloor$$

is nonincreasing as q' increases from p : the linear term drops by one while the floor increases by at most one. Its value at $q' = p$ is at most one, because

$$2p \leq 3m - 5 \implies \left\lfloor \frac{2p - 2}{m - 2} \right\rfloor \leq 2.$$

But the left side is at least $2q' + 2$, since $\lceil Dq'/(D - 1) \rceil = q' + \lceil q'/(D - 1) \rceil \geq q' + 1$. This contradiction proves the lower bound. $\square$

Theorem 16 (One nontrivial star on one side). *For all integers $a, b \geq 3$, $r \geq 1$ and $u, v \geq 0$,*

$$\widehat{r}(rK_{1,a} \sqcup uK_2, K_{1,b} \sqcup vK_2) = r(a + b - 1) + bu + av - (\min(a, b) - 1) \min(u, v).$$

Proof. If a, b have opposite parity, use Theorem 15 with $s = 1$. If both are odd, every diagonal is good, so the all-good lower bound and Lemma 4 prove the formula. It remains to consider even $a, b \geq 4$. Put

$$D = a + b - 2 \geq 6, \quad g = D - \max(a, b) = \min(a, b) - 2 \geq 2.$$

Fix a, b, u, v and choose a failing $r \geq 1$ least. As in the preceding proof, pad to $e(G) = r(D+1) + T - 1$ and obtain $\Delta(G) \leq D$: use the preceding multiplicity when $r \geq 2$, and the complete good tail when $r = 1$. Lemma 11 gives $d_D(G) \geq r$.

Let I be a maximum independent set of $C_D(G)$, of size M . We have $M \geq r$. Otherwise maximality of I would give $\Delta(G - I) \leq D - 1$, while its suffix still begins on the $D + 1$ plateau, forcing degree at least D by Lemma 3. Write $M = r + t$, $t \geq 0$.

Let S_t be the sum of the good tail after its first t entries, setting $S_t = 0$ after exhaustion. For $H = G - I$, the all-good bound, or nonnegative edge count after exhaustion, gives $\delta = e(H) - S_t \geq 0$. Exact independent-vertex deletion and the bound on each tail entry give

$$\delta = r - 1 - Dt + (T - S_t) \leq r - 1 - gt. \quad (4)$$

This remains valid after exhaustion because $T \leq ct$. In particular $t \leq \lfloor (r - 1)/g \rfloor < r$.

Let z count isolated K_{D+1} components of G . Lemma 12 gives

$$z \geq 2d_D(G) - \alpha(C_D(G)) \geq r - t.$$

Deleting I leaves z isolated K_D components in H . Its remaining good tail consists only of even entries and ones. Lemma 13 therefore gives

$$\delta \geq z(D/2 - 1) \geq 2(r - t).$$

Together with (4), this requires $r + 1 + (g - 2)t \leq 0$, impossible. There is no failing r . The upper bound again follows from Lemma 4. $\square$

These theorems allow arbitrarily long matching tails and arbitrarily many repeated nontrivial stars on one side. They do not impose a bound on the total number of components. They concern the displayed families rather than arbitrary star forests.

7 Comparison and remaining scope

The matching-tail results are not an application of Theorem 1. For example, for every $r \geq 2$ and $u, v \geq 1$, Theorem 16 gives

$$\widehat{r}(rK_{1,4} \sqcup uK_2, K_{1,4} \sqcup vK_2) = 7r + 4u + 4v - 3 \min(u, v).$$

Its first r diagonal maxima are all 7 and are not good, so the local gap condition fails. Every later diagonal is good. Thus the exact value is r edges above the specific simultaneous-suffix lower bound quoted from [10]. This is an unbounded improvement over that bound; it is not a comparison with every consequence of the earlier literature.

These examples have matching components in both forests, repeated nontrivial stars in the first forest, and even star sizes. They fall outside the uniform, all-odd, one-star, and equal-two-star

hypotheses of the theorem statements in [3], and outside the one-nontrivial-star-per-forest scope described in Cheng’s institutional abstract [5]. The final diagonal is 1, so the numerical Győri–Schelp condition as restated in [3] also does not apply. The case with one nontrivial star on each side overlaps the scope of Cheng’s thesis, and the all-odd cases were already known. These subcases are included for a uniform statement and are not claimed as new.

The current version of Fu, Luo, and Ni [6] concerns uniform star forests and retains the general formula as a conjecture. As in the earlier comparison, noncoverage by the stated hypotheses does not establish priority. The full original Győri–Schelp paper and the full Cheng thesis were not available in the literature audit.

What remains open. The results here do not establish the formula for arbitrary lists of nontrivial star sizes. Even among forests with only one repeated nontrivial size on each side and appended matchings, the opposite-parity theorem covers only its stated multiplicity range, and the both-even argument covers a single nontrivial star on one side. Leading size two with matching components on both sides is also outside the new all-parity theorem. No full resolution of Erdős problem 561 is asserted.

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